Test Plan: Group Movie Recommendation System

Project: Group Movie Recommendation system
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1. Introduction

This document outlines the testing strategy, test cases, and quality evaluation criteria for the Group Movie Recommendation System. The application allows individual users to create accounts, join one or multiple watch groups via unique invite codes, vote on short sets of candidate movies according to their current mood, and obtain a ranked group recommendation list generated across the entire catalogue. The recommendation engine applies a hybrid scoring mechanism that blends current member votes, public metadata, and historical group feedback.

The purpose of this test plan is to verify that all functional requirements (FR-01 through FR-15) and non-functional requirements (NFR-01 through NFR-07) are thoroughly verified prior to final project acceptance.

2. Test Items

The following software modules and components will undergo verification:

- **User and Group Management Module:** Handles user registration, login credential validation, group creation, unique join codes, and display name uniqueness per group.
- **Preference Collection and Session Module:** Selects batches of 10 movies spanning multiple genres, collects binary Interested or Not Interested votes, stores records securely, and enforces voting locks until all active members finish.
- **Recommendation Generation Engine:** Performs group preference aggregation with equal member weighting, scans the entire 10,000 movie catalogue, computes hybrid scores, and renders the ranked results list.
- **Feedback and Personalization Module:** Records 1 to 5 integer ratings from individual members, calculates group average ratings, and applies past feedback to influence future recommendation runs.
- **Data Persistence and Security Layer:** Validates all user inputs, ensures data integrity across system restarts, isolates group data, and handles errors without exposing private tokens or system traces.

3. Features to be Tested

4.1 Functional Features

- **FR-01 User Authentication:** Account creation and login using valid credentials, along with rejection of invalid attempts.
- **FR-02 Group Creation and Joining:** Creation of new groups, generation of unique invite codes or links, and rejection of invalid codes.
- **FR-03 Unique Display Names:** Enforcement of distinct display names among members inside the same group.
- **FR-04 Movie Selection:** Presenting each group member with a configurable selection of 10 movies across diverse genres.
- **FR-05 Preference Submission:** Recording binary Interested or Not Interested votes for each presented movie.
- **FR-06 Preference Storage:** Associating submitted preferences with exact user, group, movie, and session identifiers.
- **FR-07 Preference Locking:** Locking member submissions upon entry and withholding recommendations until every participating member has finished voting.
- **FR-08 Group Preference Aggregation:** Aggregating individual votes with equal weight across all members to establish group compatibility.
- **FR-09 Whole Catalogue Candidate Selection:** Evaluating candidate movies from the full catalogue of up to 10,000 titles, estimating affinity for unevaluated titles.
- **FR-10 Hybrid Recommendation Ranking:** Scoring candidates based on current compatibility, public ratings, and past group feedback, including graceful fallback for new groups.
- **FR-11 Recommendation Display:** Presenting top recommendations with title, synopsis, and public rating.
- **FR-12 Individual Recommendation Ratings:** Accepting member review ratings between 1 and 5.
- **FR-13 Group Feedback Aggregation:** Calculating arithmetic mean ratings for recommended movies.
- **FR-14 Historical Feedback Utilization:** Leveraging prior group ratings in subsequent sessions without changing completed historical data.
- **FR-15 Input Validation and Error Handling:** Guarding against invalid formats, ensuring database rollbacks on failure, and showing clear user messages.

4.2 Non-Functional Features

- **NFR-01 Performance:** Generating recommendations within 3 seconds for at least 95 percent of requests with up to 10 members and 10,000 catalogue items.
- **NFR-02 Data Persistence and Integrity:** Preserving records across server restarts without orphaned rows.
- **NFR-03 Authentication and Authorization:** Restricting group-specific data and actions to authenticated, authorized members.
- **NFR-04 Data Privacy:** Preventing disclosure of user passwords, tokens, or raw technical error traces.

- **NFR-05 Usability:** Ensuring at least 4 out of 5 first-time participants complete the complete flow without guidance.
- **NFR-06 Modular Design and Testability:** Maintaining decoupled modules where recommendation algorithms can be unit-tested without the user interface.
- **NFR-07 Recommendation Quality:** Confirming via offline evaluation that the hybrid algorithm achieves higher NDCG at 10 compared to single-signal baselines.

4. Features Not to be Tested

The following items are outside the scope of this testing cycle:

- **Third-Party Streaming Playback:** Direct video streaming, playback DRM, or media player integration.
- **External Social Network Login:** Third-party single sign-on providers such as Google or Facebook, as local credential management is the primary target.
- **Production Cloud Scalability:** Multi-region cloud load balancing or automated container autoscaling beyond the specified 10-member group test benchmark.

5. Testing Approach

Testing will follow a structured multi-level approach spanning unit testing, integration testing, system performance benchmarking, and usability trials.

- **Unit Testing:** Automated test suites will verify isolated modules, such as mathematical formulas for mean ratings, candidate score weighting, input validation rules, and offline metric calculation routines. In accordance with NFR-06, recommendation logic will run completely separate from the UI layer.
- **Integration Testing:** Integration runs will verify the interaction between the session manager and the voting store. A primary focus is testing voting synchronization barriers to ensure the system does not calculate recommendations until every group member has completed their submission.
- **System and Performance Testing:** The entire software stack will run on a dedicated test machine. Synthetic datasets containing 10,000 movie titles and automated virtual users will measure response times under multi-member group workloads.
- **Usability Testing:** Five first-time participants will execute the core workflow from registration to recommendation rating to observe intuitive navigation and confirm user-friendly system feedback.

6. Item Pass/Fail Criteria

The testing effort will evaluate software readiness against the following acceptance criteria:

Quality Dimension	Pass Condition	Fail Condition
Functional Correctness	All 15 functional requirements (FR-01 to FR-15) pass test execution with expected data states.	Any functional failure, premature session completion, or incorrect ranking calculation.
Response Latency	Recommendation generation completes within 3.0 seconds for at least 95 percent	More than 5 percent of recommendation queries take longer

	of test runs.	than 3.0 seconds under benchmark load.
Data Integrity	All preferences, group links, and feedback persist accurately through database restarts with zero broken references.	Any data loss, dropped feedback record, or orphaned database row detected after restart.
Security and Privacy	All protected endpoints reject unauthorized requests, and logs/UI reveal zero passwords or tokens.	Exposure of passwords, sensitive session tokens, or internal database stack traces to clients.
Usability Success	At least 4 out of 5 first-time test participants complete the end-to-end journey unassisted.	Two or more participants require intervention or cannot finish the group recommendation workflow.
Algorithmic Quality	The hybrid recommendation engine achieves superior NDCG at 10 on the test set compared to baseline models.	The hybrid engine performs equal to or worse than simple rating-only or preference-only baselines.

7. Test Cases and Traceability Matrix

The table below details explicit test cases mapped directly to project requirements.

Test ID	Reqs	Test Objective & Steps	Expected Result
TC-AUTH-01	FR-01 NFR-04	User Registration and Login Validation 1. Register new user with valid username and password. 2. Attempt login with correct password. 3. Attempt login with incorrect password.	Valid login returns authentication token. Invalid login is rejected with clear error. Password is not echoed in responses.
TC-GRP-01	FR-02 FR-03	Group Creation and Duplicate Display Name Check 1. Create group and generate join code. 2. User A joins using code. 3. User B attempts to join same group using User A's display name.	User A joins group successfully. User B's join request is rejected due to duplicate display name in group.
TC-GRP-02	FR-02 NFR-03	Invalid Group Code and Access Control 1. Attempt joining with non-existent group code. 2. Non-member attempts to read group session data.	Invalid code is rejected with helpful error. Unauthorized data request returns access denied.
TC-PREF-01	FR-04 FR-05	Movie Presentation and Voting Collection 1. Active member opens voting screen. 2. Verify batch of 10 movies across diverse genres. 3. Member marks each title as Interested or Not Interested.	System serves 10 titles. All submitted votes are recorded accurately with no unselected items permitted.
TC-SESS-01	FR-06 FR-07	Session Preference Locking and Multi-User Barrier 1. Group of 3 members starts session. 2. Members 1 and 2 submit votes. 3. Verify session state. 4. Member 3 submits votes.	Member submissions cannot be modified after entry. Recommendation generation remains blocked until Member 3 finishes.
TC-RECOM-01	FR-08 FR-09	Equal Influence Aggregation Across Full Catalogue 1. Setup conflicting member votes (e.g. Action vs Comedy). 2. Trigger recommendation over 10,000	Catalogue movies not presented during voting are evaluated based on metadata. All member inputs carry equal influence.

		catalogue items. 3. Inspect candidate selection scores.	
TC-RECOM-02	FR-10 FR-11	Hybrid Scoring and Cold-Start Fallback 1. Run recommendation for brand new group with zero past ratings. 2. Verify generated ranking and presentation format.	Engine falls back smoothly to current votes and public ratings. Results display title, overview, and rating.
TC-FB-01	FR-12 FR-13	Feedback Rating and Group Mean Computation 1. Three members rate recommended title with values 3, 4, and 5. 2. Query group aggregate summary.	Individual ratings stored properly. Group arithmetic mean is recorded in the range of 3.99 to 4.01.
TC-FB-02	FR-14	Historical Feedback Influence on Subsequent Sessions 1. Complete Session 1 and record feedback. 2. Start Session 2 for the same group. 3. Generate recommendations and verify score adjustment.	Session 2 recommendations reflect historical group feedback weights. Session 1 stored records remain unchanged.
TC-ERR-01	FR-15 NFR-04	Data Validation and Graceful Error Handling 1. Submit malformed vote payload. 2. Simulate database failure during feedback save. 3. Inspect error responses.	Malformed request rejected without corrupting data. Database rollbacks prevent phantom records. No internal stack trace exposed.
TC-PERF-01	NFR-01	Recommendation Latency Benchmark 1. Populate database with 10,000 movies. 2. Simulate 10 group members submitting simultaneously. 3. Measure elapsed time to deliver ranked list across 100 iterations.	At least 95 out of 100 runs complete within 3.0 seconds, meeting latency target.
TC-REL-01	NFR-02	Data Persistence Verification Across System Restarts 1. Populate user, group, vote, and feedback data. 2. Restart backend application and database service. 3. Run integrity audit queries.	All records persist intact. Zero orphaned records exist; foreign keys between users, groups, and sessions remain valid.
TC-USE-01	NFR-05	First-Time Usability Assessment 1. Five novice users execute complete journey independently. 2. Track completion rate and points of friction.	At least 4 out of 5 users complete account creation, group joining, voting, and result review without assistance.
TC-EVAL-01	NFR-07	Offline Recommendation Algorithm Quality Evaluation 1. Run offline benchmark script on held-out evaluation dataset. 2. Compute NDCG at 10 for hybrid engine versus baselines.	Hybrid recommendation engine achieves higher NDCG at 10 compared to rating-only and preference-only baselines.

8. Test Environment and Setup

Testing will take place in a controlled development and staging environment configured as follows:

- **Hardware and Host Environment:** Standard test workstation or virtual instance equipped with 4 CPU cores, 8 gigabytes of RAM, and fast local solid-state storage.
- **Software and Runtime:** Node.js or Python backend application runtime, modern web browser (Google Chrome or Mozilla Firefox), and relational database server (PostgreSQL or SQLite).
- **Test Movie Dataset:** Local catalogue loaded with 10,000 movie records containing complete metadata including title, synopsis, genres, and validated public average ratings.
- **Evaluation Split:** Dedicated held-out test partition used specifically to compute ranking accuracy metrics without data leakage.
- **Test Frameworks:** Automated test execution via PyTest or Jest; API and load simulation using Locust or Postman runner.

9. Roles and Responsibilities

Team assignments for the execution of this test plan are designated as follows:

Role	**Name**	**SRN**
Test Lead / QA Coordinator	Ahana H Sharan	PES1UG24AM020
Backend / Machine Learning Developer	Akshata Amara	PES1UG24AM025
Frontend / Full-Stack Developer	Anarghyaa Kashyap	PES1UG24AM039
Project Reviewer / Stakeholder	Annanya Babu	PES1UG24AM045

10. Risks and Contingencies

- **Risk 1:**
  Performance Bottleneck with 10,000 Movies
  Complex candidate similarity scoring across the entire catalogue could push response times beyond the 3-second threshold during peak loads.
  Contingency: Implement two-stage candidate selection using genre and popularity filtering before executing multi-factor ranking.
- **Risk 2:**
  Voting Race Conditions in Groups
  Concurrent vote submissions by multiple group members could lead to dirty reads or desynchronized session completion.
  Contingency: Implement atomic database updates and transaction locking around session status transitions.
- **Risk 3:**
  Cold Start in First-Time Groups
  New groups lack historical feedback, which could degrade hybrid recommendation scoring.
  Contingency: Enforce automated fallback logic (FR-10) that dynamically reallocates historical weight to current member votes and public ratings.